Vectors in $\mathbb{R}^3$: Distances, Closest Points & Reflections

Point to a Line

Line L: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$, point $\mathbf{p}$

Closest point on the line

$$\lambda_* = \frac{(\mathbf{p}-\mathbf{a})\cdot\mathbf{b}}{\mathbf{b}\cdot\mathbf{b}}, \qquad \mathbf{r} = \mathbf{a} + \lambda_*\mathbf{b}$$

Shortest distance (cross-product)

$$\text{dist} = \frac{|(\mathbf{p}-\mathbf{a})\times\mathbf{b}|}{|\mathbf{b}|}$$

Shortest distance (dot-product)

$$\text{dist}^2 = (\mathbf{p}-\mathbf{a})\cdot(\mathbf{p}-\mathbf{a}) - \frac{\big((\mathbf{p}-\mathbf{a})\cdot\mathbf{b}\big)^2}{\mathbf{b}\cdot\mathbf{b}}$$

Point to a Plane

Plane Π: $\mathbf{r}\cdot\mathbf{n} = d$, point $\mathbf{p}$

Closest point on the plane

$$\mathbf{r} = \mathbf{p} - \frac{\mathbf{p}\cdot\mathbf{n} - d}{|\mathbf{n}|^2}\,\mathbf{n}$$

Shortest distance

$$\text{dist} = \frac{|\mathbf{p}\cdot\mathbf{n} - d|}{|\mathbf{n}|}$$

Cartesian form
Π: $ax + by + cz = d$, $P = (x_0, y_0, z_0)$

$$\text{dist} = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2+b^2+c^2}}$$

Distance Between Two Lines

$$L_1\colon\ \mathbf{r} = \mathbf{a}_1 + \lambda\mathbf{b}_1, \qquad L_2\colon\ \mathbf{r} = \mathbf{a}_2 + \mu\mathbf{b}_2$$

Skew lines

$$\text{dist} = \frac{|(\mathbf{a}_2-\mathbf{a}_1)\cdot(\mathbf{b}_1\times\mathbf{b}_2)|}{|\mathbf{b}_1\times\mathbf{b}_2|}$$

Parallel lines (shared direction $\mathbf{b}$)

$$\text{dist} = \frac{|(\mathbf{a}_2-\mathbf{a}_1)\times\mathbf{b}|}{|\mathbf{b}|}$$

Closest points on skew lines — solve simultaneously:

$$\begin{cases} \big((\mathbf{a}_1+\lambda\mathbf{b}_1) - (\mathbf{a}_2+\mu\mathbf{b}_2)\big)\cdot\mathbf{b}_1 = 0 \\ \big((\mathbf{a}_1+\lambda\mathbf{b}_1) - (\mathbf{a}_2+\mu\mathbf{b}_2)\big)\cdot\mathbf{b}_2 = 0 \end{cases}$$

Line–Plane & Parallel Planes

Line meets / parallels a plane

L: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$, Π: $\mathbf{r}\cdot\mathbf{n} = d$

If $\mathbf{b}\cdot\mathbf{n} \neq 0$: the line meets the plane (dist $= 0$) at parameter λ_* given by

$$\lambda_* = \frac{d - \mathbf{a}\cdot\mathbf{n}}{\mathbf{b}\cdot\mathbf{n}}$$

If $\mathbf{b}\cdot\mathbf{n} = 0$: the line is parallel to the plane.

$$\text{dist} = \frac{|\mathbf{a}\cdot\mathbf{n} - d|}{|\mathbf{n}|}$$

Distance between parallel planes

Π_1: $\mathbf{r}\cdot\mathbf{n} = d_1$, Π_2: $\mathbf{r}\cdot\mathbf{n} = d_2$
Same normal $\mathbf{n} \implies$ parallel planes.

$$\text{dist} = \frac{|d_2 - d_1|}{|\mathbf{n}|}$$

Reflections in a Plane

Plane Π: $\mathbf{r}\cdot\mathbf{n} = d$

Reflect a point

$$\mathbf{p}' = \mathbf{p} - 2\,\frac{\mathbf{p}\cdot\mathbf{n} - d}{|\mathbf{n}|^2}\,\mathbf{n}$$

Reflect a direction vector

$$\mathbf{b}' = \mathbf{b} - 2\,\frac{\mathbf{b}\cdot\mathbf{n}}{|\mathbf{n}|^2}\,\mathbf{n}$$

Reflect an entire line L: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$

$$L'\colon\ \mathbf{r} = \mathbf{a}' + \lambda\mathbf{b}'$$

where $\mathbf{a}'$ and $\mathbf{b}'$ given by the above.

Angle Formulae

Acute angle between two lines

$$\cos\theta = \frac{|\mathbf{b}_1\cdot\mathbf{b}_2|}{|\mathbf{b}_1|\,|\mathbf{b}_2|}$$

Acute angle between two planes

$$\cos\theta = \frac{|\mathbf{n}_1\cdot\mathbf{n}_2|}{|\mathbf{n}_1|\,|\mathbf{n}_2|}$$

Acute angle between a line and a plane

$$\sin\theta = \frac{|\mathbf{b}\cdot\mathbf{n}|}{|\mathbf{b}|\,|\mathbf{n}|}$$